

CS & IT ENGINEERING



COMPILER DESIGN

PARSING

$\left\{ \begin{array}{l} LR(0) \\ SLR(1) \end{array} \right\} \underline{\underline{LR(0) DFA}}$
 $\left\{ \begin{array}{l} CLR(1) \\ LALR(1) \end{array} \right\} \underline{\underline{LR(1) DFA}}$

Lecture No.- 08



By- Venkat sir

Recap of Previous Lecture



Topic

??

Home Work

CLR(1)

① {
 $S \rightarrow Aa$
 $S \rightarrow bAc$
 $S \rightarrow Bc$
 $S \rightarrow bBa$
 $A \rightarrow d$
 $B \rightarrow d$
}



Topics to be Covered



Topic

??

Topic

CLR()

Topic

??

Q



Lookahead LR(1)

No R/R Conflict

✓ $CLR(1) = 13$

LALR(1) = 12
SLR(1) = 12

$S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot A a, \$$
 $S \rightarrow \cdot b A c, \$$
 $S \rightarrow \cdot B c, \$$
 $S \rightarrow b \cdot B a, \$$
 $A \rightarrow \cdot d, \{a\}$
 $B \rightarrow \cdot d, \{c\}$

I_0

$S \xrightarrow{s} S' \rightarrow S \cdot, \$$
 I_1

$A \xrightarrow{A} S \rightarrow A \cdot a, \$ \xrightarrow{a} S \rightarrow A a \cdot, \$$
 I_6

$B \xrightarrow{B} S \rightarrow B \cdot c, \$ \xrightarrow{c} S \rightarrow B c \cdot, \{ \$ \}$
 I_2 I_3 I_7

$b \xrightarrow{b} S \rightarrow b \cdot A c, \$ \xrightarrow{A} S \rightarrow b A \cdot c, \$ \xrightarrow{c} S \rightarrow b A c \cdot, \$$
 I_4 I_8 I_9

$S \rightarrow b \cdot B a, \$ \xrightarrow{B} S \rightarrow b B \cdot a, \$ \xrightarrow{a} S \rightarrow b B a \cdot, \$$
 I_{10} I_{11}

$A \rightarrow \cdot d, \{a\}$
 $B \rightarrow \cdot d, \{c\}$
 $d \xrightarrow{d} A \rightarrow d \cdot, \{c\}$
 $B \rightarrow d \cdot, \{a\}$
 I_{12}

$d \xrightarrow{d} A \rightarrow d \cdot, \{a\}$
 $B \rightarrow d \cdot, \{c\}$
 I_5

$A \rightarrow d \cdot, \{a, c\}$
 $B \rightarrow d \cdot, \{a, c\}$
 $I_{5 \& 12}$

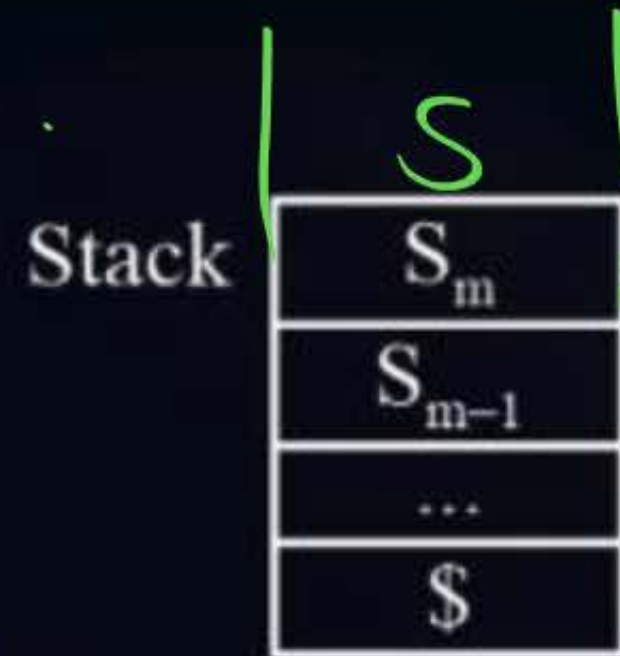
R/R Conflict



Topic : LR Parsing

Shift/Reduce Parsing

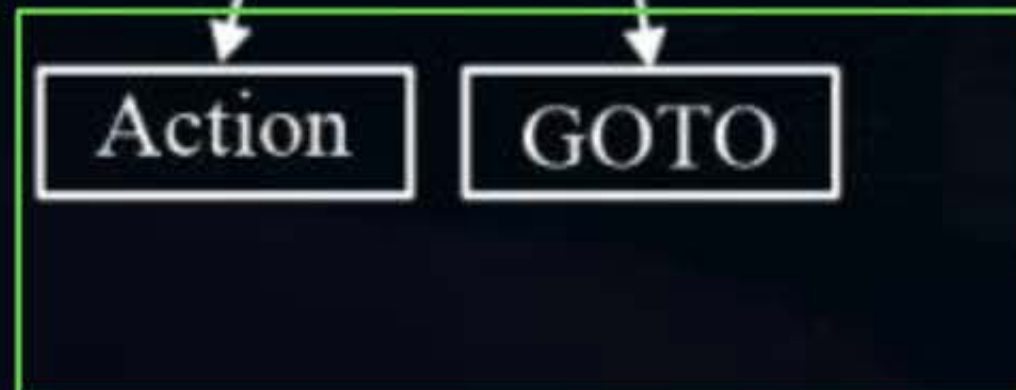
→ R.N.D in Reverse
→ Left to Right scan



Parse tree

Output

Syntax error



LR Parsing Table



Topic : The LR parsing algorithm

- At each step, the parser is in some configuration.
 - The next move depends on reading a_1 from the input and s_m from the top of the stack.
- ① If action $[s_m, a_i] = \text{shift } s$ we execute a SHIFT move, entering the configuration $(s_0 X_1 S_1 \dots X_m S_m a_i s, a_{i+1} \dots a_n \$)$.
 - ② If action $[s_m, a_i] = \text{reduce } A \rightarrow B$, then we enter the configuration $(s_0 X_1 S_1 \dots X_{m-r} S_{m-r} A s, a_{i+1} \dots a_n \$)$, where $r = |\beta|$ and $s = \text{goto } [s_{m-r}, A]$.
 - ③ If action $[s_m, a_i] = \text{accept}$, we're done.
 - ④ If action $[s_m, a_i] = \text{error}$, we call an error recovery routine.

S is state of DFA on top of stack
 a is Lookahead Symbol



① $T[S, a]$ = shift _{j}

✓ a) Push a into stack.

✓ b) Push j into top of stack

✓ c) Increment Lookahead Symbol.

$T[S, a] = \text{Reduce } [A \rightarrow B]$

~~ⓐ~~ pop $(2 * |B|)$ symbols from stack.

~~ⓑ~~ Push A into top of stack.

ⓒ push $\text{GOTO}(i, A)$ into top of stack.

i if previous state on top of stack

#Q. How many shift action and reduce action are taken by LR parser to parse input **ab** by using following grammar and parsing table .

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow b$



		action			goto		
		a	b	\$	S	A	B
I_0	S_3				1	2	
I_1				Accept			
I_2		S_5					4
I_3		r_2					
I_4			R_1				
I_5			r_3				



Topic : The LR parsing algorithm

$\begin{cases} 1. S \rightarrow AB \\ 2. A \rightarrow a \\ 3. B \rightarrow b \end{cases}$ ab

$T.C = O(n)$

Stack input action

① $\$ \mid 0 \mid$

$a \mid b \mid \$$

$T[0, a] = S_3 \checkmark$

Shift = ②
Reduce = ③

② $\$ \mid 0 \mid a \mid 3$

$b \mid \$$

$T[3, b] = r_2 \text{ (Reduce)} \text{ (A} \rightarrow a) \text{ } 2 \times 1 = 2$

③ $\$ \mid 0 \mid A \mid 2$

$b \mid \$$

$T[2, b] = S_5 \checkmark$

④ $\$ \mid 0 \mid A \mid 2 \mid b \mid 5$

$\$$

$T[5, \$] = r_3 \text{ Reduce (B} \rightarrow b) \text{ } 2 \times 1 = 2$

⑤ $\$ \mid 0 \mid A \mid 2 \mid B \mid 4$

$\$$

$T[4, \$] = r_1 (S \rightarrow AB) = 2 \times 2 = 4$

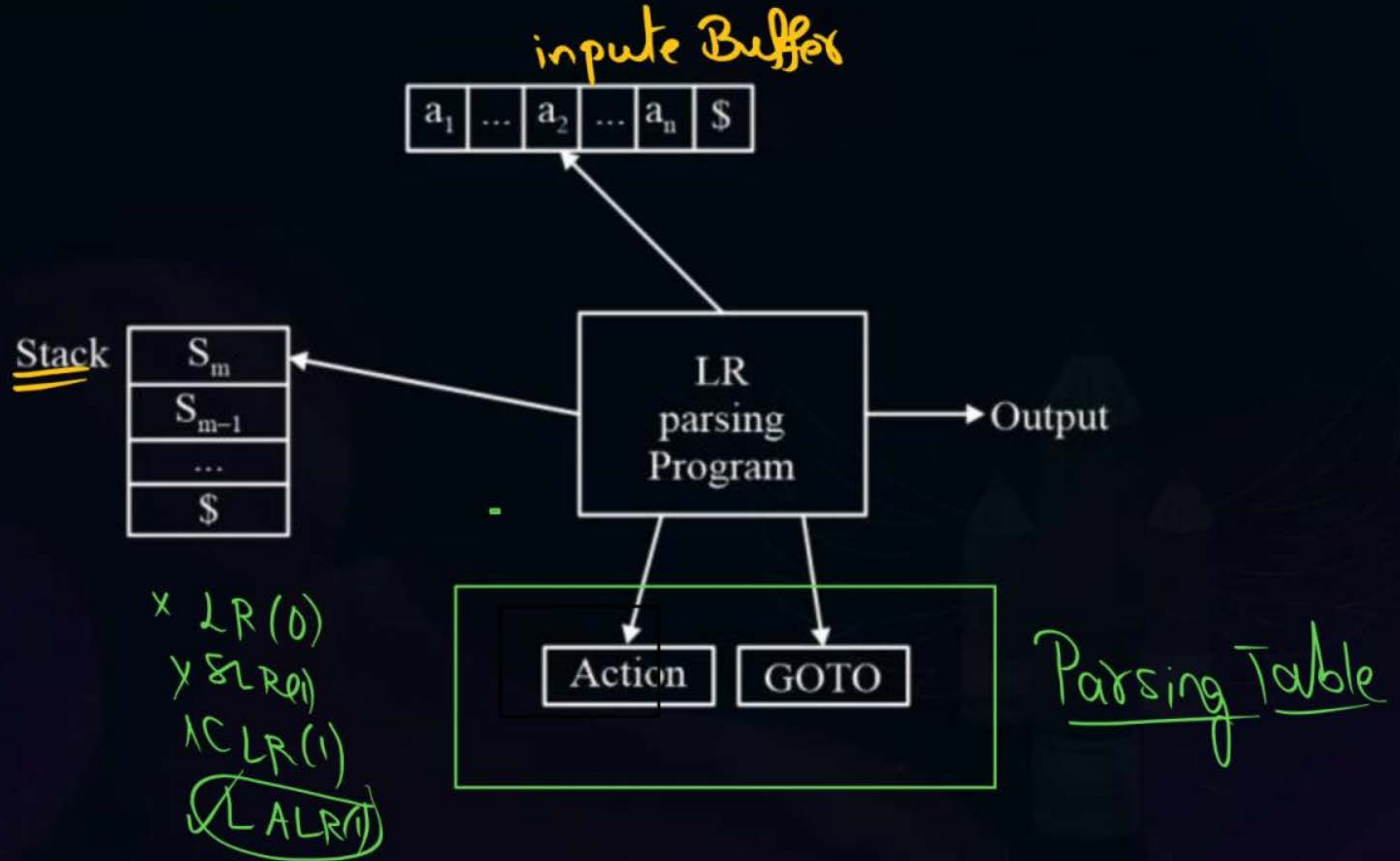
⑥ $\$ \mid 0 \mid S \mid 1$

$\$$

$T[1, \$] = \text{accepted} \checkmark$



Topic : LR Parsing





Topic : The LR parsing algorithm

Time Complexity = $O(n)$
↳ length of input

- ① At each step, the parser is in some configuration.
 - The next move depends on reading a_i from the input and s_m from the top of the stack.
- ① • If action $[s_m, a_i] = \text{shift } s$, we execute a SHIFT move, entering the configuration $(s_0 X_1 S_1 \dots X_m S_m a_i s, a_{i+1} \dots a_n \$)$.
- ② • If action $[s_m, a_i] = \text{reduce } A \rightarrow B$, then we enter the configuration $(s_0 X_1 S_1 \dots X_{m-r} S_{m-r} A s, a_{i+1} \dots a_n \$)$, where $r = |\beta|$ and $s = \text{goto } [s_{m-r}, A]$.
- ③ • If action $[s_m, a_i] = \text{accept}$, we're done.
- ④ • If action $[s_m, a_i] = \text{error}$, we call an error recovery routine.

#Q. How many shift action and reduce action are taken by LR parser to parse input **ab** by using following grammar and parsing table .

$S \rightarrow AB$

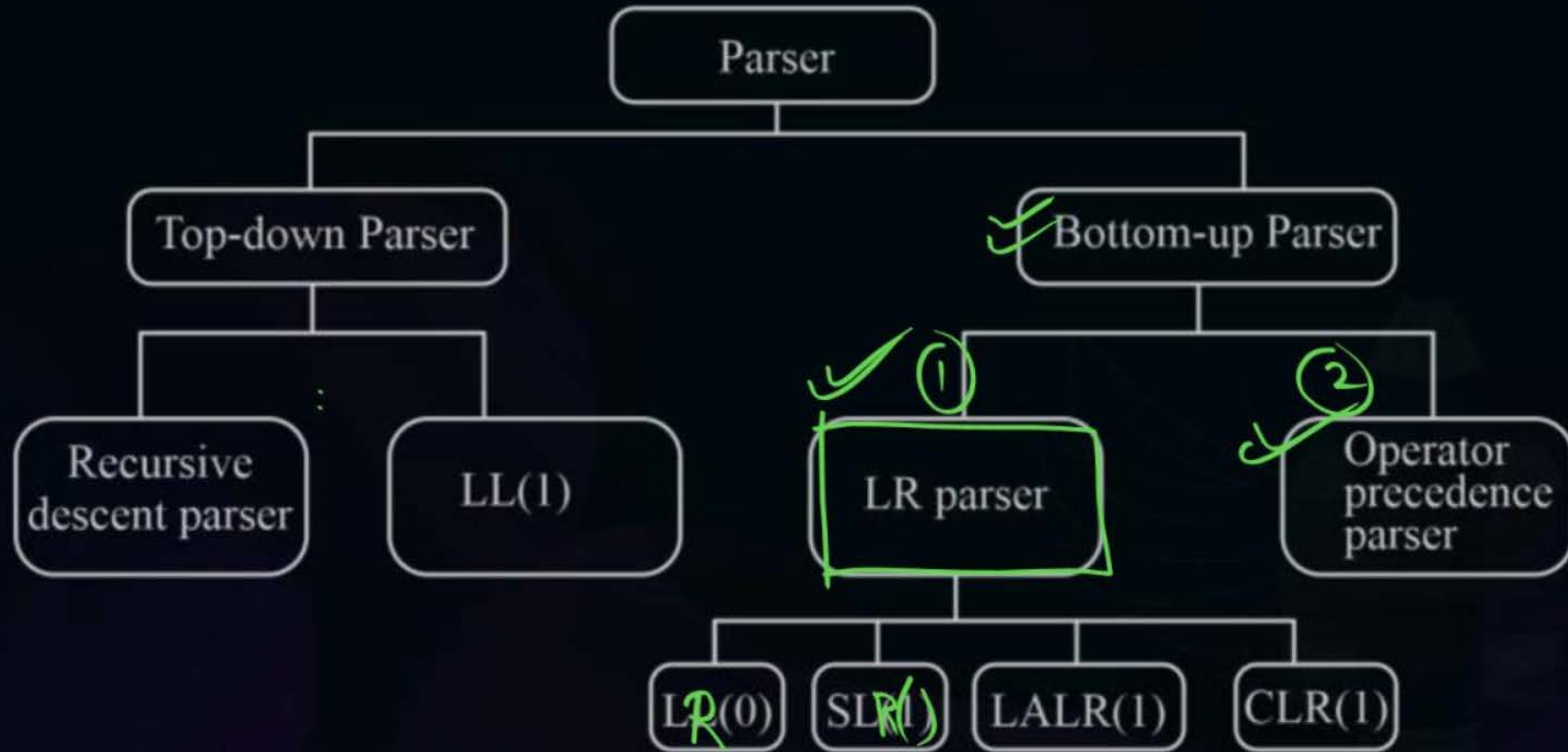
$A \rightarrow a$

$B \rightarrow b$

	A	B	\$	S	A	B
I_0	S_3			1	2	
I_1			Accept			
I_2		S_5				4
I_3		r_2				
I_4			R_1			
I_5			r_3			



Topic : PARSER





Operator Precedence Parsing

① Op. Parsing can Handle Ambiguous Grammar



Operator Grammar

(CFG)

*

+

①

No ϵ -productions

②

no adjacent non terminals on grammar R.H.S

Operator Grammar? ^{not}



① ~~✗~~ $\left\{ \begin{array}{l} S \rightarrow \underline{AB} \\ A \rightarrow a \\ B \rightarrow b \end{array} \right\}$

~~✗~~ ② $\left\{ \begin{array}{l} S \rightarrow A \\ A \rightarrow \epsilon \end{array} \right\}$

✓✓ ③ $\boxed{\begin{array}{l} S \rightarrow S * T \\ T \rightarrow a \end{array}}$

✓✓ ④ $\left\{ \begin{array}{l} E \rightarrow E + T \mid T \\ T \rightarrow T * F \mid F \\ F \rightarrow a \end{array} \right\}$

④ > ③

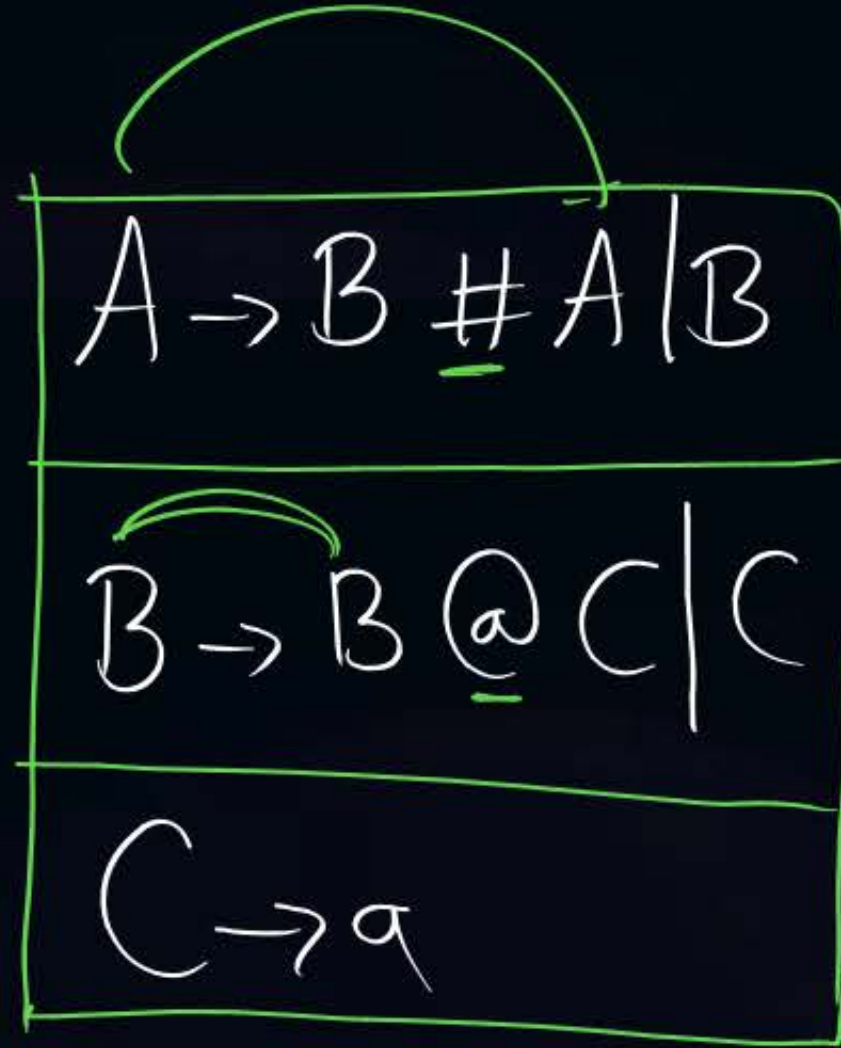
Parse tree



Right Associative

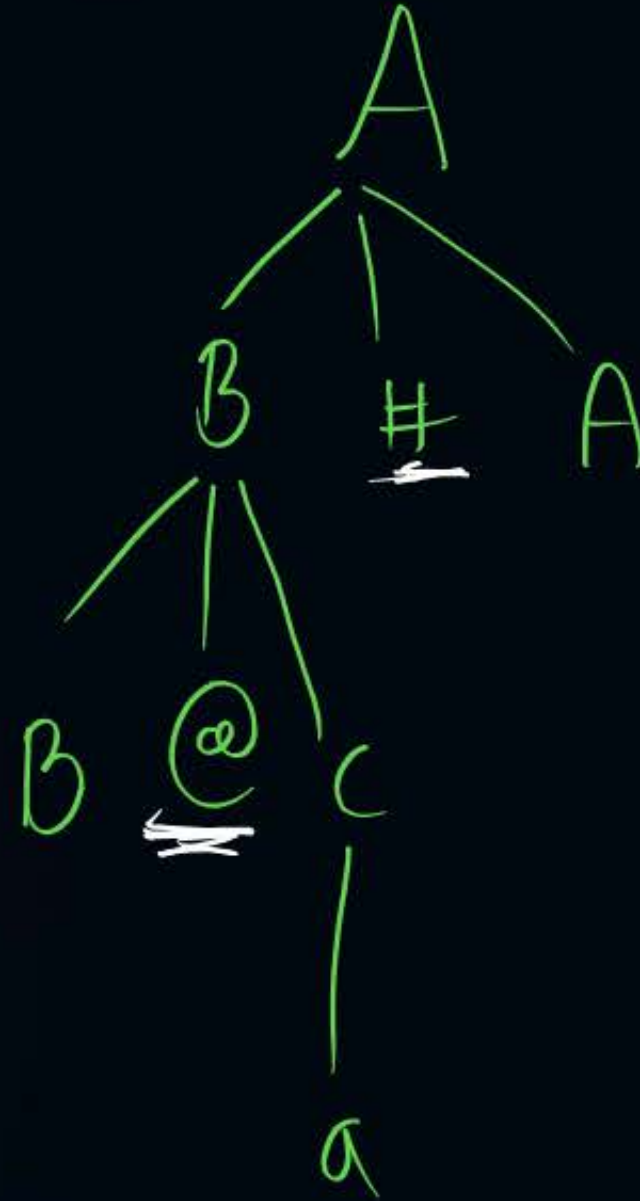


②



$\# \Rightarrow R.A$
 $@ \Rightarrow \underline{\text{Left Associative}}$

$@ > \#$



$$\textcircled{3} \quad A \rightarrow B \uparrow A \mid B$$

$$B \rightarrow B \& C \mid C$$

$$C \rightarrow D @ C \mid D$$

$$D \rightarrow a$$



$$\textcircled{a} \quad \uparrow < \&$$

$$\textcircled{b} \quad \& < @$$

$$\left\{ \begin{array}{l} @ \Rightarrow R.A \\ \& \Rightarrow L.A \\ \uparrow \Rightarrow R.A \end{array} \right.$$

$$(4) \quad A \rightarrow \underline{B \uparrow A} \mid \underline{A \& B} \mid B$$

$$B \rightarrow B \# C \mid C @ B \mid C$$

$$C \rightarrow a$$

$$(1) \quad \uparrow = \&$$

$$(2) \quad \# = @$$

$$(3) \quad \uparrow < \#$$





2 mins Summary



Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

THANK - YOU

